VTU • Department of Biotechnology

Traditional Medicine & Health Management

Subject Code: BBT755C | Model Answers to Previous Year Question Paper (Q.5 – Q.10)

Modules 3, 4 & 5 • Prepared from the module PPTs

Q.No	Topic	Module
5a	Classification of secondary plant chemicals; properties & functions of two classes	3
5b	Pharmacokinetics; drug absorption & distribution	3
6a	Toxicology of medicinal plants: side effects & contraindications	3
6b	Sensitivity to a medical substance; metabolism & excretion	3
7a	Need for Quality Assurance in herbal medicine & key processes	4
7b	Essential dispensary skills	4
8a	Legislation: sourcing, storage & labelling of herbal medicines	4
8b	Forms of administration of herbal & traditional medicine	5
9a	Padartha Vijnana, Rachana Sharira & Kriya Sharira	5
9b	Rasayana & Vajikarana – purposes	5
10a	Roga Vijnana vs Vikruthi Vijnana	5
10b	Short note: Panchakarma & Kayachikitsa	5

How to score full marks (10-mark answer):

- Write the **definition / introduction** first (2–3 lines), then **headings with points**.
- Draw the **figure / flow chart** neatly with a pencil and label it – figures carry separate marks.
- Underline **keywords** and write **examples** (e.g. Morphine, Triphala, Ashwagandha). End with a 1–2 line **conclusion**.
- Q.6, Q.8 & Q.10 are **OR** choices – attempt either Q.5 or Q.6 etc. as per the paper; answers for all are given here.

Q.5a Answer

10 Marks | Module 3 • L3 • CO1

Classify secondary plant chemicals and describe the properties and functions of any two classes.

Introduction

Secondary plant chemicals (secondary metabolites) are organic compounds produced by plants that are **not directly involved in growth, development or reproduction** (unlike primary metabolites such as carbohydrates, proteins and lipids). They act as the plant's "*defence and survival toolkit*" and are the main source of herbal drugs.

Characteristics

- Not essential for basic survival but important for **adaptation**; produced in specific parts (roots, leaves, bark, seeds, flowers).
- Occur in **small quantities**; protect the plant against herbivores, pathogens and competition (defence, signalling, ecological interaction).

Classification

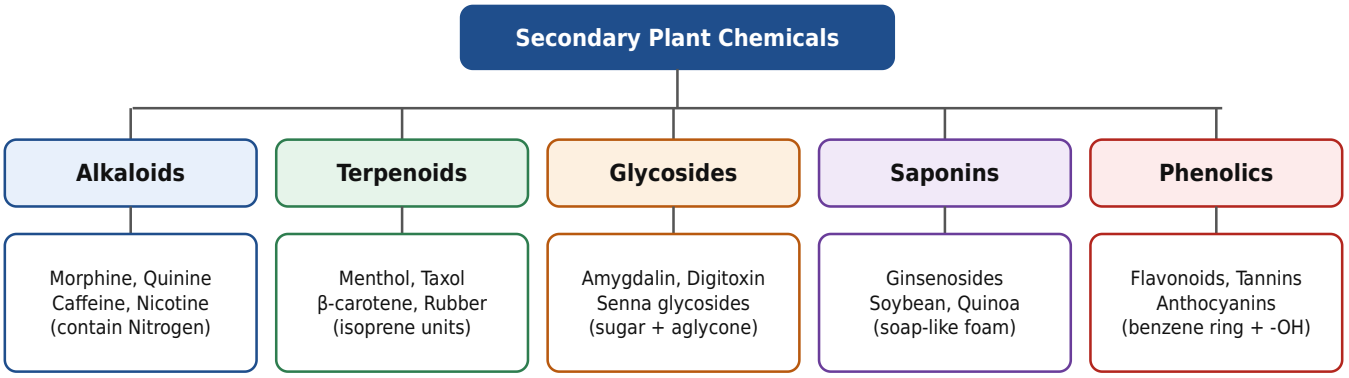


Fig. 1: Classification of secondary plant chemicals with examples

Class	Chemical nature	Examples & use
Alkaloids	Nitrogen-containing, basic compounds	Morphine (analgesic), Quinine (antimalarial)
Terpenoids	Built from 5-carbon isoprene units	Menthol, β-carotene, Taxol (anticancer)
Glycosides	Sugar (glycone) + non-sugar (aglycone)	Digitoxin (cardiac), Amygdalin (cyanogenic), Senna (laxative)
Saponins	Amphiphilic glycosides; foam in water	Ginsenosides, soybean; lowers cholesterol
Phenolics	Aromatic ring with one or more -OH groups	Flavonoids, tannins, anthocyanins; antioxidant

1. Alkaloids

Definition: Diverse **nitrogen-containing** natural compounds with a **basic (alkali-like)** nature.

- **Chemical structure:** one or more nitrogen atoms, usually in a **heterocyclic ring**; derived from amino acids (tryptophan → indole alkaloids; tyrosine → isoquinoline alkaloids); form **salts with acids**.
- **Physical properties:** mostly **crystalline solids** (nicotine is liquid); soluble in organic solvents, **alkaloid salts are water-soluble**; **bitter** taste; **toxic** in high doses.
- **Functions:** in plants – protective substances, metabolic regulators, nitrogen reserve. In humans – wide pharmacological effects:

Alkaloid	Formula	Function
Morphine	$C_{17}H_{19}NO_3$	Opioid analgesic (painkiller)
Quinine	$C_{20}H_{24}N_2O_2$	Antimalarial
Caffeine	$C_8H_{10}N_4O_2$	Central nervous system stimulant
Nicotine	$C_{10}H_{14}N_2$	Neurotoxin; natural insecticide

2. Terpenoids

Definition: Natural organic compounds built from **5-carbon isoprene units (C_5H_8)** with extra functional groups (often oxygen). Synthesised by the **mevalonate** and **MEP/DOXP** pathways.

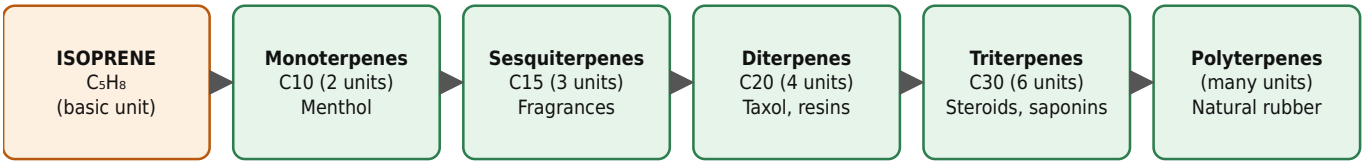


Fig. 2: Terpenoids are built from repeating 5-carbon isoprene units

- **Physical properties: volatile, hydrophobic, strongly aromatic**; many are **colour-giving pigments** (carotenoids – yellow sunflower, red tomato).
- **Functions & examples: Menthol** ($C_{10}H_{20}O$) – cooling sensation, insect repellent; **β-carotene** ($C_{40}H_{56}$) – orange pigment, **provitamin A**; **Taxol** – anticancer; **Rubber** (polyisoprene) – elasticity, commercial use. Terpenoids give scent and flavour (eucalyptus, cinnamon, ginger) and show strong antimicrobial and anti-inflammatory activity.

Conclusion: Secondary metabolites are classified by chemical nature into alkaloids, terpenoids, glycosides, saponins and phenolics. Each protects the plant and provides important medicines to humans.

Q.5b Answer

10 Marks | Module 3 • L2 • CO1

Define the term 'Pharmacokinetics'. Briefly explain the process of drug absorption and distribution in the human body.

Pharmacokinetics – Definition

Pharmacokinetics is the study of **what the body does to the drug** – i.e. the movement of a drug **into, through and out of the body**, covering **A**bsorption, **D**istribution, **M**etabolism and **E**xcretion (**ADME**). Key parameters: clearance, half-life (t½) and steady-state concentration.

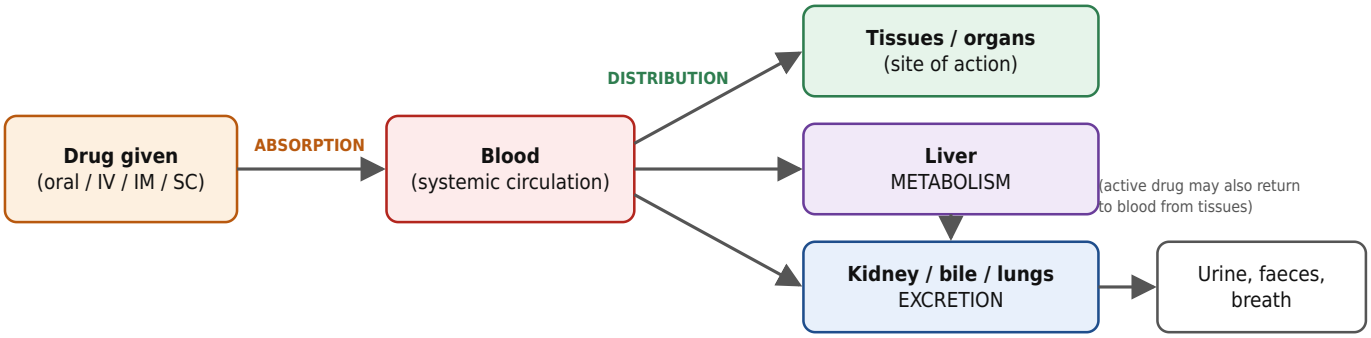


Fig. 3: ADME – the journey of a drug in the human body (Pharmacokinetics)

1. Absorption

Definition: Absorption is the process by which a drug **enters the bloodstream from its site of administration**.

Mechanisms of absorption

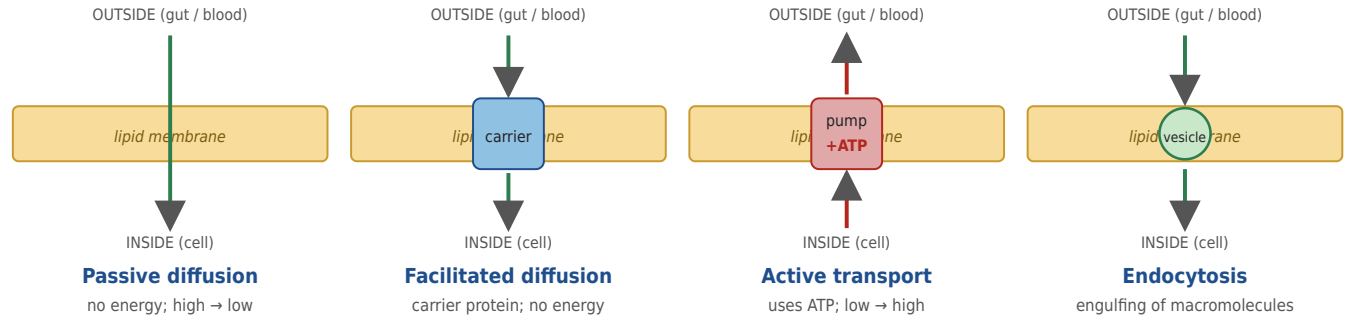


Fig. 4: Mechanisms of drug absorption across the cell membrane

- **Passive diffusion** – high → low concentration, no energy; the **most common** route (lipid-soluble drugs, e.g. diazepam).
- **Facilitated diffusion** – carrier protein, no energy (e.g. vitamin B_{12}).
- **Active transport** – energy-dependent, against gradient, saturable (e.g. levodopa).
- **Endocytosis / pinocytosis** – engulfing of macromolecules (e.g. insulin).

Factors influencing absorption

Factor	Effect / example
Physicochemical properties	Small, lipid-soluble, non-ionised drugs absorb faster; large or ionised ones (insulin) slower
Route of administration	Bioavailability: **IV (100%) > IM > SC > Oral**
pH and ionisation	Non-ionised form crosses membrane; weak acids absorb in stomach, weak bases in intestine
Surface area & blood flow	Small intestine > stomach (large surface, rich blood supply)
GI contents	Food, enzymes, gastric motility (food delays some antibiotics)
Formulation	Particle size, coating; controlled-release vs immediate-release tablets

Bioavailability (F): fraction of administered drug that reaches systemic circulation **unchanged**. IV = 100%; oral drugs are reduced by **first-pass metabolism** in the liver.

2. Distribution

Definition: The **reversible transfer** of a drug from the bloodstream to tissues and organs.

- **Blood flow** – high in heart, liver, kidney; low in fat and bone.
- **Plasma protein binding** – drug bound to albumin is **inactive**; only the **free drug** is active.
- **Tissue binding** – accumulation in tissues (tetracycline in bone).
- **Lipid solubility** – lipophilic drugs cross membranes (and barriers such as the blood–brain barrier) easily.
- **Volume of distribution (Vd)** – extent of spread: **high Vd** = widely in tissues (chloroquine); **low Vd** = confined to plasma (heparin).

> **Conclusion:** Absorption brings the drug into the blood; distribution carries it to the site of action. Both decide the onset, intensity and duration of drug action.

Q.6a Answer

10 Marks | Module 3 • L3 • CO2

Explain the toxicology of the commonly used medicinal plants, including side effects and contraindications.

Introduction

Toxicology is the study of the **adverse effects of substances** (natural or synthetic) on living organisms. Though medicinal plants are natural, many contain **potent bioactive compounds** that cause toxic reactions if used improperly. Toxicity depends on the **dose, duration, plant part used, preparation and individual sensitivity**.

Reasons for toxicity



Fig. 5: Reasons for toxicity in medicinal plants

Types of toxic effects (side effects)

Type	Plant / compound	Mechanism / effect
Hepatotoxicity	Pyrrolizidine alkaloids (comfrey); turmeric (rarely); kava	Hepatocellular necrosis, cholestasis
Nephrotoxicity	Aristolochia species, Aloe latex	Tubular necrosis
Cardiotoxicity	Digitalis (cardiac glycosides), Ephedra	Arrhythmia, hypertension
Neurotoxicity	Ginkgo (ginkgotoxin), St. John's Wort	Seizures, serotonin syndrome
Haematologic	Ginkgo, garlic, ginger	Increased bleeding tendency
Endocrine / reproductive	Ginseng, licorice	Hormonal imbalance, pseudo-aldosteronism
Dermatologic / allergic	Aloe (topical), chamomile	Contact dermatitis

Other examples: *Aconitum* and *Aristolochia* are toxic if improperly prepared; Senna overuse causes electrolyte imbalance; cyanogenic glycosides (bitter almond, cassava) release **HCN**.

Cautions and contraindications

Contraindications (do NOT use)

- During **pregnancy** and **lactation** (e.g. Senna, Aloe latex, Aristolochia).
- In **liver or kidney disease** (hepatotoxic / nephrotoxic herbs).
- Known **allergy** to the plant family (chamomile, daisy family).
- **Before surgery**: stop ginkgo, garlic, ginger (bleeding risk).
- Heart disease: Ephedra, licorice (raise blood pressure).

Cautions (precautions)

- Use the **correct species, part and dose**; avoid long-term use.
- Buy **standardised**, tested products free of heavy metals and pesticides.
- Watch for **herb–drug interactions** (St. John's Wort reduces effect of many drugs through CYP induction).
- Take **medical supervision** for toxic herbs; stop at the first allergic or adverse sign and report it (pharmacovigilance).

Conclusion: Natural does not always mean safe. Correct identification, standardised dosage, awareness of contraindications and monitoring of side effects are essential for safe use of medicinal plants.

Q.6b Answer

10 Marks | Module 3 • L2 • CO1

Explain the concept of 'sensitivity' to a medical substance. How does the body metabolize and excrete these substances?

1. Concept of sensitivity

Sensitivity is the **degree of response of an individual** to a given dose of a medicinal substance. The same dose may be effective in one person, ineffective in another and toxic in a third because people differ in sensitivity.

Type	Meaning / example
Hypersensitivity (allergy)	Immune-mediated, **not dose-related**, needs earlier exposure. Rash, itching, anaphylaxis; contact dermatitis from topical Aloe or chamomile
Idiosyncrasy	Unusual, **genetically determined** reaction on first dose
Tolerance	Reduced response on repeated doses – larger dose needed
Increased / decreased sensitivity	Exaggerated or weak response compared with the average person

Factors affecting sensitivity: genetics (slow / fast acetylators), **age** (newborns and elderly metabolise slowly), **liver or kidney disease**, pregnancy, enzyme induction / inhibition by other drugs, diet and environment.

2. Metabolism (Biotransformation)

Definition: Enzymatic conversion of drugs into **more water-soluble compounds** so they can be excreted easily in urine or faeces. **Main site: liver** (cytochrome P450 system); also kidney, lungs, intestine and skin.

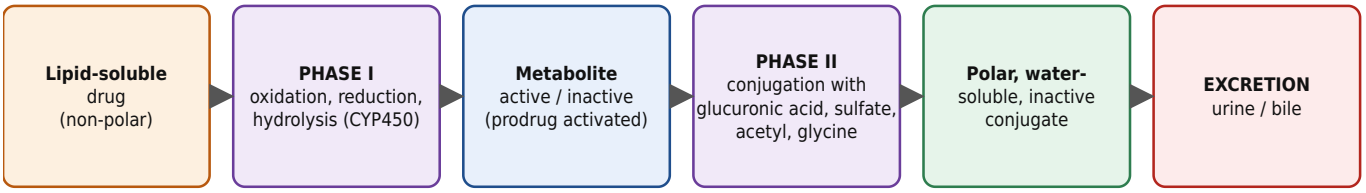


Fig. 6: Biotransformation (metabolism) of a drug in the liver

Phase	Reaction	Purpose & example
Phase I (functionalisation)	Oxidation, reduction, hydrolysis by **CYP450** enzymes	Exposes groups (-OH, -NH_2, -SH). May activate, e.g. **codeine → morphine**, or form a **prodrug** (sulfasalazine); diazepam → oxazepam
Phase II (conjugation)	Glucuronidation, sulfation, acetylation, methylation, glutathione	Drug + endogenous molecule → **polar, inactive** product, e.g. morphine glucuronide, paracetamol sulfate, isoniazid (acetylation)

CYP450: CYP3A4, CYP2D6, CYP2C9 are the key isoenzymes. **Inducers** (rifampicin, phenobarbital) ↑ metabolism → ↓ efficacy; **inhibitors** (erythromycin, ketoconazole) ↓ metabolism → ↑ toxicity.

3. Excretion

Definition: Elimination of the drug or its metabolites from the body.

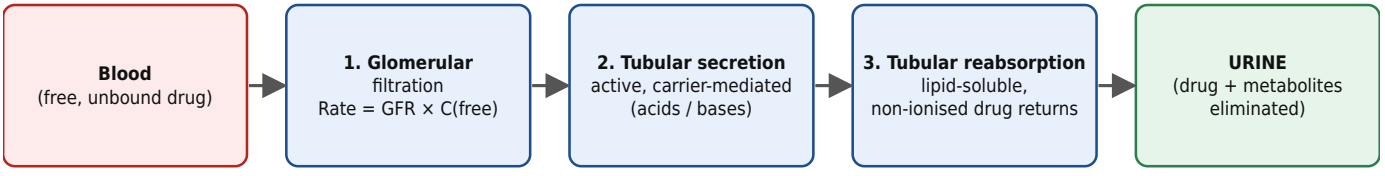


Fig. 7: Steps of renal excretion

- **Renal excretion (main route):** glomerular filtration (free drug only), tubular secretion (e.g. penicillin) and tubular reabsorption (lipid-soluble drugs). **Ion trapping:** to eliminate a weak acid (salicylate) **alkalinise urine** with $NaHCO_3$ → drug ionised → less reabsorption → faster excretion.
- **Biliary excretion:** large polar conjugates → bile → intestine → faeces.
- **Pulmonary excretion:** volatile substances (anaesthetic gases). Minor routes: sweat, saliva, milk.

Conclusion: Sensitivity decides how strongly a person responds to a substance, while metabolism (Phase I and II) and excretion convert it into water-soluble form and remove it from the body.

Q.7a Answer

10 Marks | Module 4 • L3 • CO2

Discuss the need for quality assurance critical in the production of herbal medicine. Outline the key process involved.

Introduction

Quality Assurance (QA) is a **systematic process** that ensures a product is developed and manufactured according to **established quality standards**. In herbal medicine it ensures the **right plant** is used, the material is **pure and standardised**, free from **contamination and adulteration**, and the **therapeutic effect is consistent in every batch**.

Need for quality assurance

- **Variation in natural sources** – species, plant part, soil, climate, harvest time and processing change the chemical content.
- **No uniform production methods** and often **no standard regulatory framework**.
- **Adulteration / contamination** – e.g. Ashwagandha roots adulterated with *Solanum nigrum* roots (reduced efficacy); heavy metals, pesticides, microbes, aflatoxins.
- Rising global use and consumer belief in safety – unsafe products (kava banned for liver toxicity) cause harm.
- To establish **identity, purity, safety and efficacy**, maintain uniformity through SOPs, give **consumer confidence** and legal compliance, and enable global acceptance.

Key processes in quality assurance

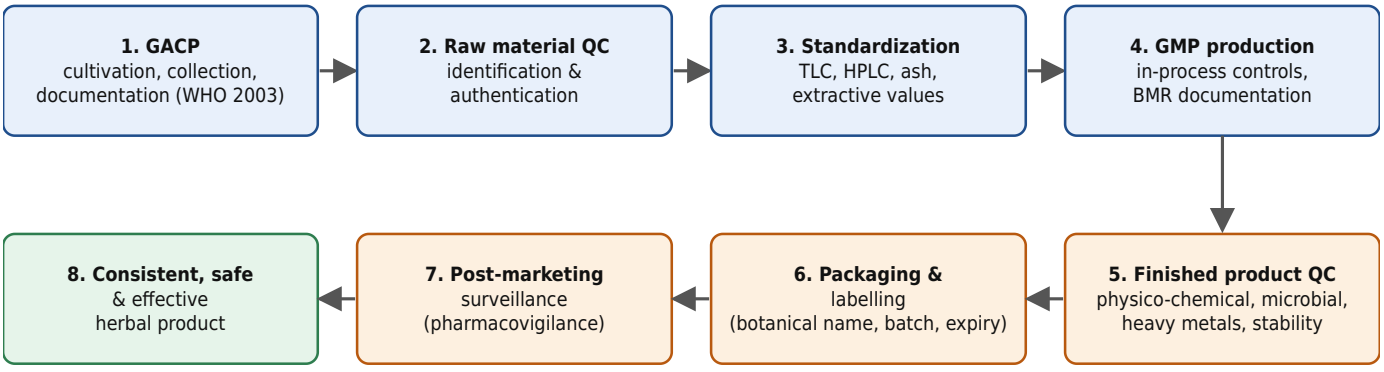


Fig. 8: Key processes of Quality Assurance in herbal medicine

Step	Details
1. Raw material QC – identification	**Macroscopic** (shape, colour, odour), **microscopic** (trichomes, stomata, xylem), **DNA barcoding**, and **voucher specimen** kept in a herbarium
2. Standardization	**Phytochemical:** TLC, HPLC, GC-MS, UV-Vis for marker compounds (curcumin in turmeric). **Physicochemical:** moisture, ash values (total, acid-insoluble, water-soluble), extractive values, pH, foreign matter. **Organoleptic:** colour, odour, taste, texture
3. GMP	Controlled environment, validated equipment, trained and hygienic personnel, **in-process controls**; documents: Batch Manufacturing Record, Master Formula Record, cleaning logs, QC reports
4. Finished product QC	Weight variation, disintegration, pH; **microbial limits** (E. coli, Salmonella, S. aureus, TAMC, TYMC); **heavy metals** (Pb, Hg, As, Cd); pesticides and aflatoxins; **stability** testing for shelf-life
5. Packaging & labelling	Non-reactive packing; label with botanical name and part used, batch no., mfg/expiry date, storage, dosage and warnings
6. Post-marketing surveillance	Monitor adverse effects and herb–drug interactions; update safety information

Standards followed: WHO Quality Control Methods (1998), WHO GACP (2003), WHO GMP for herbal medicines (2007); Indian Pharmacopoeia, Ayurvedic Pharmacopoeia of India, Chinese Pharmacopoeia, USP.

Conclusion: QA at every stage – from cultivation to the patient – ensures that herbal medicines are authentic, pure, safe and consistently effective.

Q.7b Answer

10 Marks | Module 4 • L2 • CO2

Describe the essential 'Dispensary skills' required for handling and preparing traditional medicines.

Introduction

Dispensary skills are the **knowledge, techniques and professional practices** required to **accurately prepare, package, label and provide** herbal or traditional medicines to patients.

Purpose: safe and effective use, quality-hygiene-accuracy in dispensing, and patient understanding and compliance.
Role of dispenser: interpret prescriptions, compound, counsel, keep records, report ADRs.

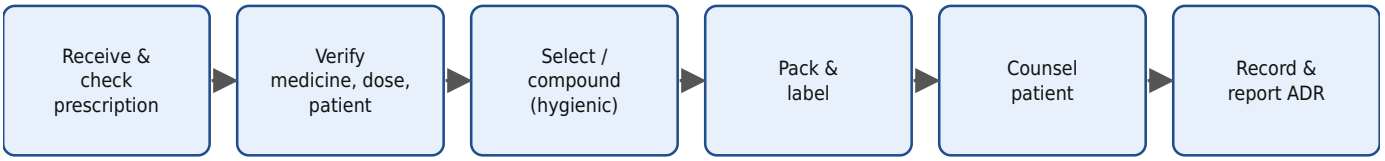


Fig. 9: Steps in dispensing a traditional medicine

Essential dispensary skills

1. Knowledge-based skills

- Understanding **formulations and dosage forms** (powders, decoctions, oils, syrups, pills, pastes).
- **Pharmacognosy** – identification, morphology and purity of medicinal plants.
- Familiarity with **Ayurvedic / Unani / Siddha / TCM pharmacopoeias** and dose standards.
- Awareness of **toxic herbs, contraindications and herb–drug interactions**.

2. Communication and counselling skills

- Explain use of medicine in **simple language**; stress **dose adherence and side effects**.
- Advise on **diet compatibility** (many traditional systems prescribe food rules).
- Encourage the patient to **report adverse effects** or unusual symptoms.

3. Technical and practical skills

- **Accurate weighing and measuring**; correct compounding and preparation of the formulation.
- Proper **packaging, labelling and storage** (cool, dry place); use of clean, calibrated tools.

4. Ethical and legal responsibilities

- **Confidentiality** of patient and respect for traditional knowledge; **accuracy** – exact measurement, **no substitution** of herbs.
- **Compliance** with Drugs & Cosmetics Act 1940 & Rules 1945; dispense only **licensed, approved** formulations; no false therapeutic claims.
- **Pharmacovigilance** – report ADRs to the AYUSH Pharmacovigilance Programme.

Good Dispensing Practice (GDP) – the six rights

Principle	Description
Right Medicine	Name and formulation match the prescription
Right Dose	Correct quantity and concentration measured
Right Patient	Confirm identity of patient
Right Time	Give frequency and timing instructions
Right Information	Educate on use, precautions and storage
Right Documentation	Maintain dispensing records and batch details

Record keeping, hygiene and patient education

- **Records:** prescription register, stock register, batch preparation / issue logs and waste-disposal records – gives **traceability** for recalls and audits.
- **Hygiene:** clean utensils and counters, **wash hands** before and after preparation, avoid cross-contamination, sterilise equipment, pest control.
- **Patient education and follow-up:** dose schedule, home storage, diet and lifestyle advice, signs of allergy; encourage feedback and follow-up visits.

> **Conclusion:** A skilled dispenser links the practitioner's prescription to the patient – ensuring the correct, safe and legally compliant use of traditional medicines.

Q.8a Answer

10 Marks | Module 4 • L3 • CO2

Discuss the legislation governing the sourcing, storage and labeling of herbal medicines.

Introduction

Legislation protects the public by ensuring that herbal medicines are **authentic, safe, traceable and truthfully labelled**. In India, they are governed mainly by the **Drugs and Cosmetics Act, 1940 and Rules, 1945** (Ayurveda, Siddha, Unani drugs – Chapter IVA, Rules 151–170) under the **Ministry of AYUSH and CDSCO**.

Region	Major legislation	Authority
India	Drugs & Cosmetics Act 1940 & Rules 1945	Ministry of AYUSH, CDSCO
WHO	Guidelines for assessment of herbal medicines (1991); GMP for herbal medicines (2007)	World Health Organization
EU	Directive 2004/24/EC (Traditional Herbal Medicinal Products)	EMA
USA	Dietary Supplement Health & Education Act (DSHEA 1994)	US FDA
China	Law on Traditional Chinese Medicine (2016)	National Admin. of TCM
Australia	Therapeutic Goods Act 1989	TGA

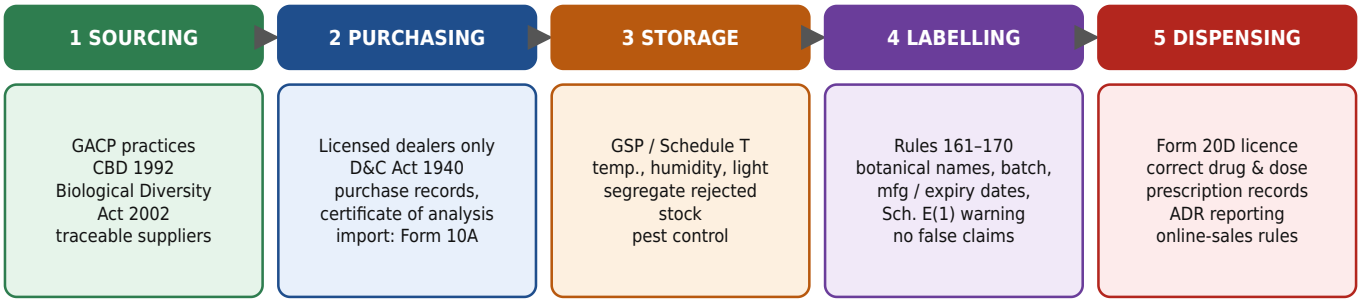


Fig. 10: Legislation at each stage of the herbal medicine supply chain

Stage 1 – Sourcing

- **Source verification:** raw material only from **registered, traceable, ethical** suppliers.
- **GACP** (Good Agricultural & Collection Practices): controlled cultivation and harvest, avoid pesticide and heavy-metal contamination, record **botanical name, location and date**.
- **Biodiversity laws:** Convention on Biological Diversity (CBD, 1992); India's **Biological Diversity Act, 2002** prohibits unauthorised collection of endemic species.
- *Objective:* authenticity, sustainability and purity.

Stage 2 – Purchasing

- Only **licensed manufacturers / dealers** may buy and sell; keep purchase records, batch numbers, supplier details and **certificates of analysis**.
- Testing for microbes, adulterants, heavy metals, aflatoxins as per AYUSH Pharmacopoeial standards; imports need **Form 10A and licence (Rule 158B)**.

Stage 3 – Storage

- **Good Storage Practices (GSP):** control **temperature, humidity and light** as per **Schedule T**.
- **Segregate** raw materials, processed goods and rejected items; containers carry **name, batch no., date of receipt and expiry date**.
- Prevent pests, cross-contamination and microbial growth. *Objective:* preserve chemical integrity.

Stage 4 – Labelling (Rules 161–170)

- **Mandatory:** name of medicine (Ayurvedic / Siddha / Unani); **ingredients with botanical names**; **batch no.** and **manufacturing licence no.**; **dates of manufacture and expiry**; dosage form, strength and directions; **manufacturer's name and address**.
- **Warning:** if toxic herbs (**Schedule E(1)**) are present – *"To be taken under medical supervision"*.
- **No false or misleading claims** (Section 33EEC); language: English, Hindi or regional language.

Dispensing: retailers need a valid **Form 20D licence**; keep prescription records; report ADRs to AYUSH Pharmacovigilance.

Penalties (India)

Violation	Section	Penalty
Misbranding / adulteration	33-I	Fine or imprisonment up to 3 years
Manufacture without licence	33-E	Imprisonment up to 1 year or fine
False therapeutic claims	33-EEC	Fine and product seizure
Unsafe / contaminated product	33-EE	Licence suspension or cancellation

Conclusion: Laws governing sourcing, storage and labelling ensure traceability, quality and patient safety at every stage of the herbal medicine chain.

Q.8b Answer

10 Marks | Module 4/5 • L2 • CO2

Enumerate and explain the various forms of administration for herbal and traditional medicine.

Introduction

Herbal and traditional medicines are given by **internal and external routes**, chosen according to the **dosha imbalance, disease stage, age, digestive power, lifestyle and therapeutic goal**. Each form has its own pharmacokinetic behaviour, bioavailability and indication.

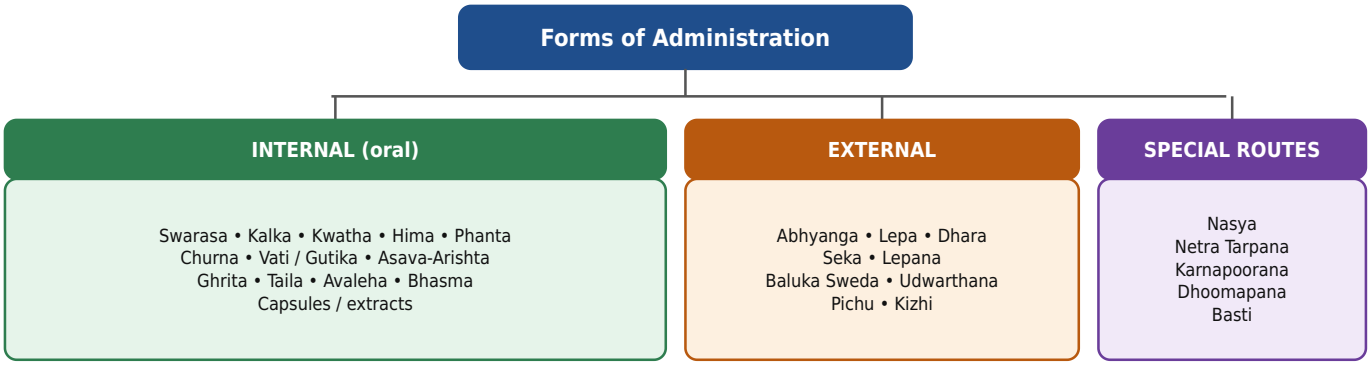


Fig. 11: Classification of forms of administration

1. Internal (oral) administration – *Ahara marga*

Form	Description	Example / use
Swarasa	Fresh juice squeezed from fresh parts; **most potent, quick absorption**	Tulasi swarasa (cold), Vasaka (cough)
Kalka	Semi-solid wet **paste**; used internally, in medicated ghee, poultices	Neem kalka (skin)
Kwatha / Kashaya	**Decoction**: coarse powder boiled in water to 1/4th volume; extracts alkaloids, tannins, glycosides	Dashamoola kwatha
Hima	**Cold infusion**: soaked overnight in cold water	Dhanyaka hima
Phanta	**Hot infusion**: hot water poured on herbs, steeped	Tulsi phanta
Churna	Finely sieved **powder**; good shelf-life, taken with water, honey or ghee	Triphala churna
Vati / Gutika	**Tablets / pills** with binders; convenient, consistent dose	Chandraprabha vati
Asava / Arishta	**Fermented** preparations with self-generated alcohol (5–12%); rapid absorption, good preservation	Drakshasava, Dashamoolarishta
Ghrita	**Medicated ghee**; improves absorption of fat-soluble compounds; neural, pitta disorders	Brahmi ghrita
Taila	**Medicated oil** (internal); vata disorders, dryness	Sahacharadi taila
Avaleha / Lehyam	Jam-like **confection** (herbs + jaggery + ghee + honey); palatable, long-term use	Chyawanprash
Bhasma	**Calcinated minerals / metals**, ultra-fine ash; tiny dose, high potency	Swarna bhasma
Capsules	Standardised herbal extracts (modern forms)	Curcumin capsules

2. External administration

Form	Use
Abhyanga	Medicated oil massage – improves circulation, reduces vata, relieves pain
Lepa	Fresh herbal paste applied locally – swelling, inflammation, burns
Parisheka / Dhara	Continuous pouring of decoction, buttermilk or oil – e.g. **Shirodhara** for mental calmness
Seka / Pralepa	Sprinkling or smearing – wounds, ulcers, skin diseases
Lepana	Poultice / warm compress – increases blood flow, relieves pain
Baluka Sweda	Sand fomentation – joint pain, stiffness
Udwarthana	Dry herbal powder massage – obesity, kapha disorders
Pichu / Kizhi	Oil-soaked cotton pad kept on head, joints, spine / heated herbal bolus applied on the body

3. Special routes

- **Nasya** – nasal instillation (sinusitis, migraine). **Netra Tarpana** – medicated ghee retained around the eyes (dryness, strain).
- **Karnapoorana** – warm oil poured in the ear. **Dhoomapana** – inhalation of medicated smoke (respiratory disorders).
- **Basti** – medicated enema (*Anuvasana* – oil; *Niruha* – decoction); the most important therapy for vata disorders.

> **Conclusion:** Choosing the correct form of administration improves absorption, effectiveness and patient compliance in traditional medicine.

Q.9a Answer

10 Marks | Module 5 • L2 • CO2

Explain the fundamental principles of Ayurveda: Padartha Vijnana, Rachana and Kriya Sharira.

1. Padartha Vijnana

Padartha means "the meaning of a word / the object that can be known". It is the Ayurvedic (Vaisheshika) philosophy of the **categories of everything that exists**. Charaka accepts **six** padarthas; Vaisheshika adds **Abhava** as the seventh.

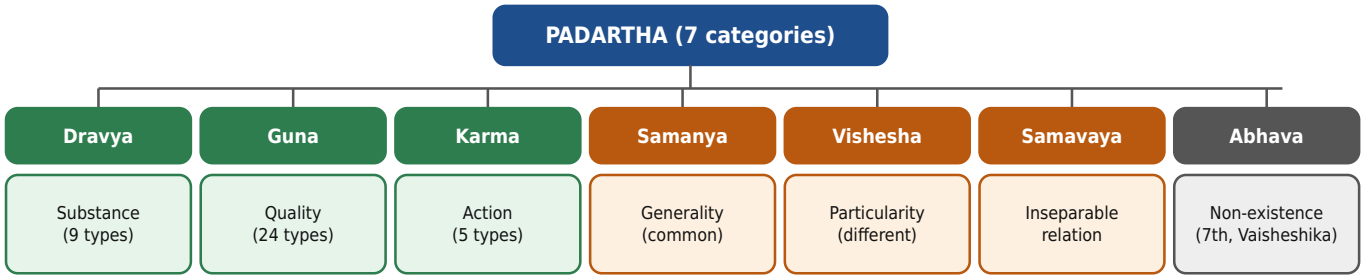


Fig. 12: Classification of Padartha (green = Dravya–Guna–Karma, the core three)

Padartha	Definition	Example
Dravya	Substance – the substrate that possesses Guna and Karma	Earth, Water
Guna	Quality that resides in dravya but has no motion	Colour, taste
Karma	Motion / action residing in dravya	Walking, expansion
Samanya	Generality common to many entities	Pot-ness
Vishesha	Particularity that differentiates entities	Individuality of atoms
Samavaya	Inseparable relation between entities	Threads and cloth
Abhava	Non-existence / absence	Absence of pot on floor

- **9 Dravya:** Prithvi (earth – smell; stability), Ap (water – taste; cohesion), Tejas (fire – form; digestion), Vayu (air – touch; movement), Akasha (ether – sound; space), plus **Kala** (time), **Disha** (direction), **Atman** (soul), **Manas** (mind).
- **24 Guna:** *Sensory (5)* Shabda, Sparsha, Rupa, Rasa, Gandha; *Quantitative / relational* Sankhya, Parimana, Prithaktva, Samyoga, Vibhaga, Paratva, Aparatva; *Psychic (6)* Buddhi, Sukha, Dukha, Iccha, Dvesha, Prayatna; *Physical (3)* Gurutva, Dravatva, Sneha; *Moral / other (3)* Dharma, Adharma, Samskara.
- **5 Karma:** Utkshepana (upward), Avakshepana (downward), Akunchana (contraction), Prasarana (expansion), Gamana (locomotion).

2. Rachana Sharira (Ayurvedic anatomy)

The branch dealing with the **structural organisation** of the human body – gross and subtle anatomy (Sources: Charaka Samhita Sharira Sthana, Sushruta Samhita, Ashtanga Hridaya). The body = **physical + physiological + spiritual** components. **Types of Purusha:** Panchabhautika (five elements), Shaddhatuka (five elements + Atma), Chaturvimshati-tattvatmaka (Sankhya, 24 principles).

Mahabhuta	Contribution to structure
Prithvi	Solidity – bones, muscles
Jala	Fluids, plasma, lubrication
Tejas	Metabolic tissues, colour
Vayu	Channels, movement
Akasha	Cavities, pores, spaces

Sapta Dhatu (7 tissues): Rasa (plasma), Rakta (blood), Mamsa (muscle), Meda (fat), Asthi (bone), Majja (marrow), Shukra (reproductive).
Tri Mala: Purisha (faeces), Mutra (urine), Sweda (sweat).

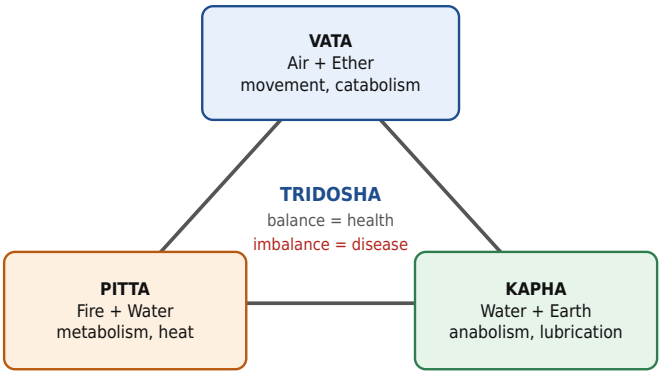


Fig. 13: Tridosha and their Mahabhuta basis

3. Kriya Sharira (Ayurvedic physiology)

Deals with the **normal functioning** of the body. Health = **balance of Dosha, Agni, Dhatu and Mala** with a pleasant mind, senses and soul.

- **Tridosha: Vata** – movement, cellular transport, electrolyte balance, elimination, **catabolism** (increased by dryness); **Pitta** – **metabolism**, body temperature, vision, hunger and thirst (aggravated by heat); **Kapha** – **anabolism**, joint lubrication (increased by sweet, fatty food).
- **Agni:** biological fire for all metabolism – 13 types; **Jatharagni** (digestive fire) is the most important. Weak agni → ulcers, diarrhoea, constipation.
- **Dhatu:** seven tissues working together – Rakta circulates blood, Mamsa supports Meda, Asthi forms bone, Majja lubricates bone, Shukra serves reproduction.
- **Mala:** waste products – failure of removal causes disease (urine retention → UTI; sweat blockage → skin problems).

> **Conclusion:** Padartha Vijnana gives the philosophical base, Rachana Sharira the structure and Kriya Sharira the function – together they explain health and disease as balance or imbalance of Dosha, Dhatu, Mala and Agni.

Q.9b Answer

10 Marks | Module 5 • L1 • CO2

What are Rasayana and Vajikarana in Ayurvedic Internal Medicine? Briefly describe their purposes.

Both are among the **eight branches of Ashtanga Ayurveda** (Kayachikitsa, Bhutavidya, Kaumarabhritya, Rasayana, Vajikarana, Shalya, Shalakya, Agada Tantra).

1. Rasayana (rejuvenation therapy)

Definition: The branch concerned with **rejuvenation, longevity, immunity and promotion of physical and mental strength**. *"Rasadi dhatunam paripurnatam kurvanam rasayanam"* – that which nourishes Rasa and the other Dhatus optimally. It is also called geriatrics in modern terms.

Purposes (objectives)

- Promotion of **longevity** (Dirghayu).
- Enhancement of **immunity** (Vyadhikshamatva).
- Improvement of **memory and intellect** (Medha).
- **Delay of ageing** (Jara nivarana).
- **Strengthening of body and mind**; prepares body after Panchakarma; prevents disease.

Rasayana drug	Property
Amalaki	Best **Vayasthapana** (anti-ageing)
Ashwagandha	Strength and immunity
Guduchi	Immunomodulator
Haritaki	Detoxification and rejuvenation
Chyavanaprasha	Compound Rasayana

2. Vajikarana (aphrodisiac / reproductive health therapy)

Definition: The branch dealing with **promotion of sexual health, fertility, virility and healthy progeny**. *"Vajivat shakti jananam vajikaranam"* – that which gives strength like a horse.

Purposes (objectives)

- Enhancement of **sexual vigour** and improvement of semen quality (**Shukra dhatu**).
- Promotion of **fertility**; production of **healthy offspring**; maintenance of reproductive health.

Indications: infertility, erectile dysfunction, premature ejaculation, low libido, general debility.

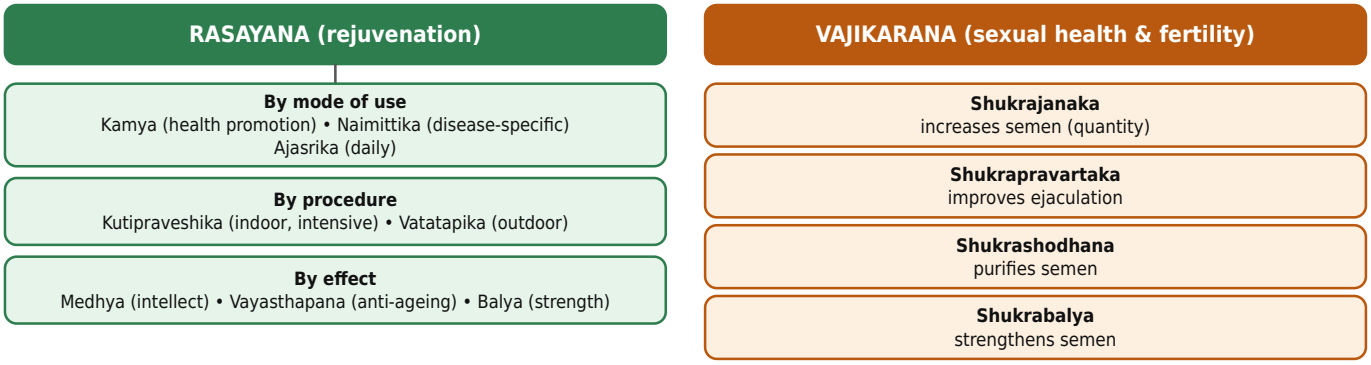


Fig. 14: Classification of Rasayana and Vajikarana

Conclusion: Rasayana rejuvenates the whole body and mind, while Vajikarana specifically improves reproductive health and progeny; both promote positive health rather than only treat disease.

Q.10a Answer

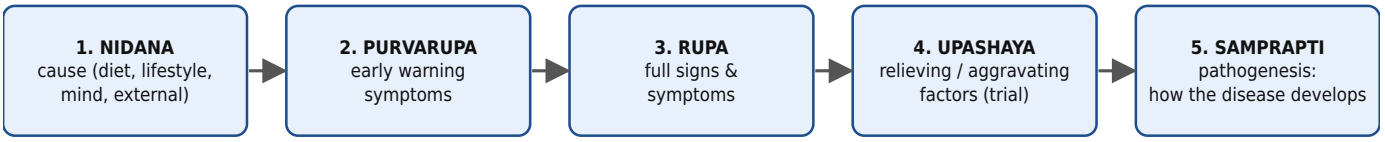
10 Marks | Module 5 • L2 • CO2

Differentiate between Roga Vijnana and Vikruthi Vijnana in Ayurvedic pathology.

Introduction

Roga Vijnana is the branch that deals with **knowledge and diagnosis of disease**: causative factors (Nidana), pathogenesis (Samprapti), clinical features (Lakshana) and prognosis (Sadhya–Asadhya). *Roga* = that which causes pain or discomfort to body and mind.

Vikruti Vijnana is the science of understanding the **abnormal (diseased) state** of body and mind by analysing **Dosha, Dhatu, Mala, Agni and Srotas**. *Vi* = abnormal, *Kruti* = state → deviation from **Prakruti** (natural constitution).



Nidana Panchaka = five tools of diagnosis in Roga Vijnana

Fig. 15: Nidana Panchaka (Roga Vijnana)

Differences

Basis	Roga Vijnana	Vikruti Vijnana
Meaning	Knowledge of **disease as an entity**	Knowledge of the **abnormal state** (deviation from Prakruti)
Definition	*Nidana–Purvarupa–Rupa–Upashaya–Samprapti jnanam Roga Vijnanam*	Study of abnormality of Dosha, Dhatu, Mala, Agni, Srotas
Focus	Diagnosis – what disease is present, how it arose and how it will progress	Understanding **which body factor** is disturbed and to what extent
Main tools	**Nidana Panchaka** (5 tools of diagnosis)	Dosha vikruti, Dhatu kshaya / vriddhi, Mala and Agni examination
Causes	Nidana: Aharaja, Viharaja, Manasika, Agantuka	Vikruti Hetu: same four groups (diet, lifestyle, mind, external)
Approach	Disease-oriented (general)	Patient-oriented, **individualised** (based on Prakruti)
Uses	Accurate diagnosis, drug selection, prevention, **prognosis**	Early detection (prevents chronicity), choice of therapy, assess curability, personalised treatment

Nidana (causes) – common to both

- **Aharaja** – improper diet (incompatible food, excess, untimely meals). **Viharaja** – lifestyle (lack of sleep, overexertion, sedentary).
- **Manasika** – stress, anger, fear, grief. **Agantuka** – injury, infection, environment.

Dosha Vikruti

Dosha	Features
Vata	Dryness, pain, anxiety
Pitta	Burning, inflammation, anger
Kapha	Heaviness, lethargy, congestion
Sannipata	Mixed symptoms

Dhatu Kshaya (depletion) / Vriddhi (excess)

Dhatu	Kshaya	Vriddhi
Rasa	Dryness, fatigue	Heaviness
Rakta	Pallor, anaemia	Bleeding
Mamsa	Muscle wasting	Growths
Meda	Emaciation	Obesity
Asthi / Majja / Shukra	Bone pain / dizziness / infertility	–

Conclusion: Roga Vijnana identifies the disease through Nidana Panchaka, whereas Vikruti Vijnana explains the disturbed body factors; together they guide accurate and personalised Ayurvedic treatment.

Q.10b Answer

10 Marks | Module 5 • L1 • CO2

Write a short note on: Panchakarma and Kayachikitsa.

1. Panchakarma

Definition: The specialised branch of Ayurveda for **systematic purification of the body by eliminating vitiated Doshas** through specific procedures. *"Doshanam mulato nirharanam Panchakarma."*

Objectives: eliminate aggravated doshas, restore **Agni**, remove **Ama** (toxins), purify **Srotas** (channels), prepare body for Rasayana, prevent and cure disease.

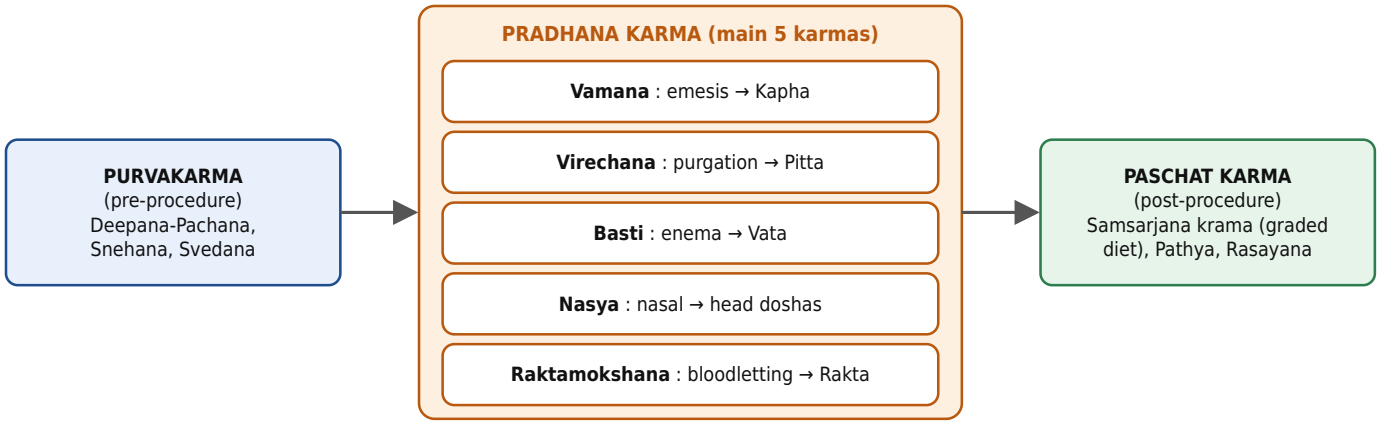


Fig. 16: The three stages and five procedures of Panchakarma

Karma	Dosha removed / route	Details
Vamana	Kapha – oral	Therapeutic emesis (decoction of liquorice, honey). Indications: kaphaja disorders, bronchial asthma, obesity, early diabetes. **Contraindicated** in pregnancy, cardiac disease, elderly
Virechana	Pitta – anal	Therapeutic purgation (senna, castor oil, psyllium). Indications: liver and skin disease, hyperacidity, fever
Basti	Vata – rectal	Medicated enema: *Anuvasana* (oil), *Niruha* (decoction); best for vata disorders
Nasya	Head doshas – nasal	Medicines (oils, decoctions, powders) through nose; sinusitis, migraine
Raktamokshana	Vitiated Rakta – bloodletting	Detoxification of blood; skin and blood disorders

- **Purvakarma: Deepana-Pachana** (improve Agni, digest Ama); **Snehana** (oleation, internal with ghee / external massage); **Svedana** (sudation by steam – liquefies doshas and eases elimination).
- **Paschat karma:** graded diet (Samsarjana krama) and regimen to restore digestion and absorption; then **Rasayana**.

2. Kayachikitsa

Definition: The branch of Ayurveda dealing with **diagnosis and management of internal diseases**. *"Antarendriya samshrita roganam chikitsa Kayachikitsa."* It is the **backbone of clinical Ayurveda**, treating disease at Dosha, Dhatu, Agni, Srotas and Manas levels.

- **Scope:** gastro-intestinal, metabolic, respiratory and cardiovascular diseases; fever, anaemia, diabetes, arthritis; psychosomatic disorders.
- **Important diseases: Jvara** (fever – king of diseases; vataja, pittaja, kaphaja, sannipataja), **Prameha** (diabetes – 20 types, kapha predominant), **Pandu** (anaemia – pitta predominant), **Grahani** (malabsorption – Agni dysfunction), **Amavata** (rheumatoid arthritis – Ama + Vata).
- **Treatment principles: Shodhana** (purification – Panchakarma), **Shamana** (palliative), **Rasayana**, **Satvavajaya** (mind control / counselling), correct diet and lifestyle.
- **Preventive aspect:** lifestyle correction, dietary regulation, seasonal regimen and Rasayana (Chyavanaprasha, Ashwagandha, Amalaki) – boosts immunity and delays ageing.

Conclusion: Panchakarma removes the root cause (vitiated doshas) by purification, while Kayachikitsa provides the complete plan for treating internal diseases; Panchakarma is the main Shodhana tool of Kayachikitsa.